

International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE, Final Term Examination, Autumn 2025
Course Code: CSE-1121 **Course Title:** Computer Programming I

Total Marks: 30

Time: 90 minutes

Figures in the right hand margin indicate full marks.

Answer all of the following questions.

[XYZ are the last three digits of your student ID.]

For example, if your ID is **C253176**, then **X = 1, Y = 7** and **Z = 6**]

1. (a) Write down the different types of data types in C programming. Explain what they are used for. **2**
(b) Consider the following variable declarations and initial values in a C program **3**

`int a = 5, b = 10, c = 3;`

`double x = 0.1, y = 0.5;`

Show how each of the following expressions will be calculated step by step using the precedence and associativity rules.

An example has been solved for you:

$a - b\%c$
 $= a - \frac{b}{c}$
 $= \frac{a - 1}{1}$
 $= 4$

- i) $a/b + x * y - (c \% b)$
ii) $(a > b) \&\& (x < y) || y + c$
(c) Write the output of the following C code segments. If there is any error in the code, rewrite it correctly. **2**

i) `int distance = X, time = Y;`
`double speed = distance/ time;`
`printf("%lf\n", speed);`

ii) `double res = Y % 1. 5;`
`printf("%lf\n", res);`

–OR–

What are pre- increment and post-increment operators in C? Explain with an examples. Why are they used?

- (d) Write a program that converts a given number of days into months (assuming each month has 30 days) and the remaining days. **3**

Sample Input	Sample Output
265	Months = 8, Days = 25
1000	Months = 33, Days = 10
16	Months = 0, Days = 16

2. (a) Design a C program for a Train Ticket Booking System. The program will ask for the type of seat and the number of tickets, then print a receipt with the total price. **4**

The ticket price for all types for adults is 400. And students get a 50% discount on the same price.

Input format: A single line containing three integers:

- Ticket type (**1 for Shovon, 2 for AC, 3 for Snigdha**)
- Number of adult tickets
- Number of student tickets

Output format: The program should display the ticket type and the total price.

Sample Input	Sample Output
2 3 2	Seat: AC Total Fare: 1600
1 2 3	Seat: Shovon Total Fare: 1400

For the first sample, Seat type = 2 (AC). And cost = 3 adults * 400 + 2 students * 200 = 1600

–OR–

Write a C program that reads a single character from User and determines its type.

- If the character is an alphabet, the program should identify whether it is an uppercase or lowercase. For uppercase letters, it should further check if the character is a vowel (A, E, I, O, U) or a consonant, and display “Uppercase Vowel” or “Uppercase Consonant” accordingly. For lowercase letters, it should perform the same check and display “Lowercase Vowel” or “Lowercase Consonant”.
- If the character is a digit (0 – 9), the program should determine whether it is even or odd and print “Even Digit” or “Odd Digit”.
- For any other character that is neither a letter nor a digit, the program should display “Special Character”.

Sample Input	Sample Output
E	Uppercase Vowel
5	Odd Digit
@	Special Character

(b) Write a C program to check if a given date is valid or invalid. The program will take three integers as input: day, month, and year, respectively.

3

A date will be considered **valid** only if it satisfies all of the following conditions:

- The year is between 1700 and 2500.
- The month is between 1 and 12.
- The day is valid for the given month:
 - i) If the month is 2 (February), the day must be between 1 and 28
 - ii) For all other months, the day must be between 1 and 30.

The program should print "Valid" if all the above conditions are satisfied. Otherwise, it should print "Invalid".

Sample Input	Sample Output
30 5 2025	Valid
31 1 2020	Invalid
29 2 2024	Invalid

(c)

3

Consider the code snippet on the right. What will be the values of x and y when n takes the following values: (i) 0 (ii) 5 (iii) 9 Explain your answer with rough calculations.	<pre>int x = 4; int y = XY % 10; If (n >= 0) if(n == 5) x -= 4, y += 10; else x *= 2, y -= 5; printf("x = %d y = %d", x, y);</pre>
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3. (a)

1

Consider the code snippet of a for loop on the right, and write equivalent programs using a (i) while loop (ii) do-while loop.	<pre>int i, sum = 0; for (i = 0; i < 5; i++) { sum += i; printf("%d %d\n", i, sum); }</pre>
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(b) Write the output of the following code segments:

2

```
i) int n = 2YZ;
   while (n)
   {
       printf("%d, ", n%10);
       n /= 10;
   }
```

```
(ii) int i = 1;
     while(i<=100)
     {
         if(i * i > XY)
             break;
         printf("%d\n", i);
         i ++;
     }
```

(c) You are given two C programs that find the square root of a given number n. Analyze the programs and determine which one is more efficient and finishes earlier. Explain your answer.

3

Code A	Code B
<pre>int n = 1024; int i = 0; while (i*i<= n) { i++; } printf("Square root of %d ≈ %d\n", n, i - 1);</pre>	<pre>int n = 1024; int low = 0, high = n, mid; while ((high - low) > 1) { mid = (low + high) / 2; if (mid * mid > n) high = mid; else low = mid; } printf("Square root of %d ≈ %d\n", n, low);</pre>

(d) You are given an initial salary. Each year, the salary increases by 5% of the current salary. Write a C program to find how many years are required for the salary to become at least 2000.

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Sample Input	Sample Output
1000	15
1500	6

For the first sample , initial salary = 1000

- After 1st year = 1000 + 5% of 1000 =1050.00
- After 2nd year = 1050 + 5% of 1050 =1102.50
-
- After 15th year = 1979.73 + 5% of 1979.93 = 2078.93

–OR–

Write a C program to check whether a given integer is an Armstrong number.

[An Armstrong number is a number that is equal to the sum of its own digits, each raised to the power of the number of the]

Example: 153 is an Armstrong number because $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

Sample Input	Sample Output
9474	Armstrong number
123	Not an Armstrong Number

Prepared by:

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